

Assignment 1 — Inspecting a New Dataset

Pandas Basics · Introduction to Machine Learning · Dataset: Titanic

Everything we did in class was on `tips`. This assignment uses a completely different dataset. **The columns change — the method does not.** That is the whole point of the exercise.

Submit one notebook file named `RollNumber_Assignment1.ipynb`.

Setup

```
import pandas as pd
import seaborn as sns

t = sns.load_dataset('titanic')      # or: t = pd.read_csv('titanic.csv')
```

Questions

- Q1** Load the dataset into `t`. How many rows and how many columns does it have?
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- Q2** Print all the column names. Pick any three and write in a comment what you think each one holds.
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- Q3** Display the first 7 rows and the last 3 rows.
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- Q4** Run `t.info()`. Some columns have fewer values than the total number of rows. **Write down which columns those are, and how many values are missing from each.** ★
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- Q5** Run `t.describe()`. It shows fewer columns than `info()` listed. Write down two columns it left out, and explain why.
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- Q6** For the `class` column, answer all three questions below using a **different function for each**:
- a) how many different values are there?
 - b) what are they?
 - c) how many passengers fall into each one?
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- Q7** Find how many passengers were male and how many were female.
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- Q8** In `describe()`, look at the age column. Compare its mean with its 50% value. Which is larger, and what does that suggest about the passengers' ages? ★
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- Q9** **(Challenge)** The age column has missing values. Using only `shape` and `info()`, work out what **percentage** of age values are missing. Write the number in a comment.
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If `sns.load_dataset` **does not work** — it needs an internet connection. Use the `titanic.csv` file instead and load it with `pd.read_csv('titanic.csv')`, keeping the file in the same folder as your notebook.